

Responsible AI basics

Module 1 - Lesson 3

Overview

Responsible AI means building systems that are fair, transparent, private, and accountable. Ethics are a design requirement, not an afterthought.

Key Concepts

- Bias: models learn biases present in training data; audit for disparate impact.
- Fairness: define who is affected and how, and measure outcomes across groups.
- Transparency: document data, decisions, and limitations; explain model behavior.
- Privacy: minimize data collection, anonymize, and honor data rights.
- Accountability: designate humans responsible for outcomes and escalations.
- Safety: test adversarial cases and monitor for harm after deployment.

Example

```
Bias check example:
model = load_model("hiring")
for group in ["women", "men", "minority", "majority"]:
    score = model.predict(sample_by_group(group))
    print(group, score.mean())
# Compare acceptance rates across groups before shipping
```

Summary

Responsible AI balances capability with fairness, transparency, privacy, and accountability. Audit data, document limits, and keep humans responsible for outcomes.

Practice Checklist

- Document who your model affects and what could go wrong.
- Run a simple bias check across demographic groups in your data.
- Write a model card summarizing data, metric, and limitations.